

A
Capstone Project Report

On

Resum.AI

Submitted during 6th semester in partial fulfilment of the requirements for the
award of degree of

Bachelor of Technology

in

Computer Engineering (with specialization in data science)

by

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CANDIDATE’S DECLARATION

I hereby certify that the work which is being presented in this Project titled “_____” in fulfilment of the requirements for the degree of Bachelor of Technology in Computer engineering and submitted to “J.C. Bose University of Science and Technology, YMCA, Faridabad”, is an authentic record of my own work carried out under the supervision of

Dr. Sonali Gupta.

The work contained in this project has not been submitted to any other University or Institute for the award of any degree or diploma by me.

Student’s Signature

Student’s Name

CERTIFICATE

It is hereby certified that the project titled _____ submitted by _____ (**Student Name and Roll no**) _____ of _____ Year (____ Semester) to “**J. C. Bose University of Science and Technology, YMCA, Faridabad**” for the award of the degree of Bachelor of Technology in Computer Engineering is a record of a bonafide work carried out by him/her in the Project Lab – J.C. Bose University of Science and Technology, YMCA, Faridabad under my supervision as mentor for May 2024 examination.

In my opinion, the project has reached the standards of fulfilling the requirements of the regulations to the degree.

Mentor Name (Mentor)

Designation

Department of Computer Engg.

ACKNOWLEDGEMENT

I/We have taken efforts in this project. However, it would not have been possible without the kind support and help of many individuals.

I would like to extend my sincere thanks to all them. I am highly indebted to my mentor _____ for their guidance and constant supervision as well as for providing necessary information regarding the project & also for their support in completing the project.

I would like to express my gratitude towards my parents & members of J.C. Bose University of Science and Technology for their kind cooperation and encouragement which helped in the completion of this project. I would like to express my special gratitude and thanks to _____ as well for their guidance.

Student's Signature

Student's Name

Brief of the Project

Recruitment is one of the most critical functions of any organization, and the resume is often the first step in assessing a candidate's suitability for a role. However, the traditional manual approach to resume screening is time-consuming, inconsistent, and error-prone. With the increasing volume of applications received for every job posting, there is a pressing need for intelligent automation in resume screening.

resum.AI is a smart resume analyzer designed to automatically evaluate resumes using artificial intelligence. The platform leverages **Google Gemini API** to understand and interpret resume content contextually and match it against job descriptions provided by recruiters or job seekers. The frontend is built using **Streamlit**, ensuring ease of deployment and user interactivity, while **SQLite3** is used as the backend database to store resume data and analysis results.

The tool not only scores resumes based on job compatibility but also provides personalized feedback for improvement. This makes **resum.AI** useful for both recruiters (to identify top talent quickly) and job seekers (to optimize resumes for better job matching).

Technologies Used

Technology	Purpose
Streamlit	Web application frontend for user interaction
Google Gemini API	Natural Language Processing and AI analysis of resumes
SQLite3	Lightweight database for storing user inputs, results, and history
Python	Backend scripting, resume parsing, and logic implementation
PDF/DOCX Libraries (e.g., PyMuPDF, python-docx)	Extract content from uploaded resume files
Regular Expressions	Content filtering and keyword matching
Matplotlib/Plotly	Optional: For visual representation of matching scores (future enhancement)

Literature Survey

While optional, it's useful to provide a survey of existing solutions and research.

2.1 Existing Resume Analyzers

Many platforms today provide resume feedback tools, such as:

- **VMock** – Offers AI-driven resume reviews with suggestions.
- **Rezi** – Focuses on keyword optimization based on job descriptions.
- **Jobscan** – Matches resume against job listings using a keyword engine.

These tools are primarily rule-based and lack contextual understanding of the content. Moreover, they don't always provide intelligent suggestions based on real semantic analysis.

Research in NLP for Resume Parsing

Natural Language Processing (NLP) research in resume parsing has grown with advances in machine learning. Transformer models like **BERT**, **GPT**, and **Gemini** offer deeper semantic understanding, making it feasible to analyze resumes beyond keyword matching. These models can:

- Recognize context and relationships in text.
- Understand nuanced differences between skill levels.
- Provide personalized suggestions.

Implementation and Result Analysis

Problem Formulation

Problem Statement:

How can we automate and improve the process of evaluating resumes in a way that is fast, unbiased, and aligned with the expectations of recruiters and hiring managers?

Challenges include:

- Parsing varied formats of resumes (PDF, DOCX).
 - Understanding content contextually (not just keywords).
 - Matching resumes against custom job descriptions.
 - Delivering actionable feedback.
 - Maintaining a simple and interactive UI.
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Objectives

- To develop an AI-based system for analyzing and scoring resumes.
 - To provide a match score between a resume and job description.
 - To highlight missing or relevant keywords and skills.
 - To deliver improvement suggestions for job seekers.
 - To store analysis results for future reference or tracking.
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Methodology**A. Resume Upload and Parsing**

- User uploads a resume file (PDF or DOCX).
- Content is extracted using Python libraries (like `fitz` for PDFs and `docx` for DOCX).
- Extracted content is preprocessed to remove noise (special characters, formatting issues).

B. Job Description Input

- User inputs a job description or pastes text from a job listing.
- The system considers this as the benchmark against which resumes are compared.

C. Integration with Google Gemini API

- Resume text and job description are sent to Gemini.
- Gemini is prompted to:

- Identify key qualifications, skills, and experiences.
- Compare resume against job criteria.
- Provide a compatibility score and reasoning.
- Suggest missing elements or improvements.

Prompt Sample for Gemini:

```
python
CopyEdit
prompt = f"""
Compare this resume with the following job description and provide:
- Match percentage.
- Key strengths and gaps.
- Recommendations for improvement.

Resume: {resume_text}
Job Description: {job_description}
"""
```

D. Database Integration (SQLite3)

- User uploads and analysis results are stored in SQLite3.
- Enables resume history tracking.
- Ensures that the same user can recheck improvements over time.

E. Frontend with Streamlit

- Simple and interactive UI for uploading resumes and viewing results.
 - Real-time interaction with AI results.
 - Clean visualization of scores, suggestions, and strengths.
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Applications and Scope

4.1 Applications

- **Recruiters:** Quickly filter resumes that align best with job descriptions.
- **HR Teams:** Reduce time and effort in the first level of screening.

- **Job Seekers:** Improve their resume to meet industry/job-specific expectations.
 - **Colleges/Universities:** Career services can use this tool to guide students in resume building.
 - **ATS Systems:** Can be integrated as an AI screening module.
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Future Scope

- **Real-time Job Matching:** Automatically match resumes to live job postings using web scraping/APIs.
 - **Multilingual Support:** Analyze resumes in multiple languages using LLMs.
 - **Gamified Resume Builder:** A module that guides users step-by-step to build better resumes.
 - **Enhanced Visual Feedback:** Charts, graphs, and timelines to show progress.
 - **Custom Model Training:** Fine-tune models with industry-specific job descriptions.
 - **Voice Assistant Integration:** Conversational AI to give verbal resume feedback.
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References

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3. SQLite Official Website: <https://sqlite.org>
4. NLP Research: “A Survey on Resume Parsing Techniques using NLP” – IJERT
5. Jobscan Blog: <https://www.jobscan.co/blog>
6. VMock Resume Feedback Platform: <https://www.vmock.com>

Screenshots :

